ZOMATO – Average Order Value (AOV) Performance Analysis

Comprehensive Business Assessment Using Excel & Tableau

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1. Executive Summary

Zomato's platform shows strong transaction volume (147,061 orders) and total revenue of ₹96.4 Cr during the observed period (May 2018 – May 2020). However, revenue quality and engagement depth reveal structural imbalances.

Key observations:

- Revenue is highly right-skewed — 80.7% of transactions fall under ₹5,000
- Top 10% of transactions contribute 78.7% of total revenue.
- 76% of customers placed only 1–2 orders.
- $R^2 = 0.003$ between AOV and order volume (no meaningful correlation).
- Student segment dominates order volume (78,301 orders).
- Income does not consistently explain AOV differences across cities.

Conclusion:

While platform activity is healthy, revenue concentration, shallow repeat behaviour, and heavy reliance on specific customer segments indicate monetisation and retention inefficiencies.

2. Business Context & Objective

Leadership identified that high order volume does not always translate into strong revenue contribution. Some cities and restaurants generate frequent low-value orders, while others show inconsistent demand patterns.

This analysis aims to:

- Identify underperforming AOV segments
- Evaluate structural drivers behind revenue imbalance
- Assess customer engagement depth
- Determine whether income, occupation, or concentration patterns explain inefficiencies
- Provide data-backed, actionable recommendations

3. Data Preparation & Validation (Excel – Power Query & Data Model)

Tools Used: Excel (Power Query, Data Model, Pivot Tables)

3.1 Data Model Structure

- Fact Table: Orders (148,668 records)
- Dimension Tables: Users (100,000), Restaurants (148,542)
- Star schema validated
- One-to-many relationships confirmed
- No orphan foreign key records

Data Sources Summary

Table Name	Type	Rows Loaded	Key Columns	Status
Orders	Fact	1,48,668	order_date, user_id, r_id, sales_amount	Loaded
Users	Dimension	1,00,000	user_id, age, gender, income	Loaded
Restaurants	Dimension	1,48,542	r_id, location	Loaded
Food	Dimension	-	f_id	Connection Only
Menu	Dimension	-	menu_id	Connection Only

- Used for potential extended dimensional analysis (not loaded to model).
- Total records across all loaded tables: 397,210

3.2 Data Cleaning & Professional Handling

Identified Issues:

- 2 non-INR transactions (USD)
- 1,609 zero-value transactions
- 2 negative-value rows
- Sales value anomaly (-1)
- No exchange rate reference available

Professional Decision:

- USD transactions excluded (no conversion reference, only 2 rows, immaterial impact)
- Sales Amount > 0 filter applied for AOV analysis
- Zero-revenue orders excluded from AOV (retained in total revenue computation where appropriate)
- 2 negative rows removed

Financial impact of exclusions:

- ₹748 (0.000075% of revenue) - immaterial
- This is proper analytical documentation - not silent deletion.

Data Preparation & Validation Steps

- Data imported and transformed using Power Query
- Verified key column integrity (user_id, r_id,)
- Checked for duplicate records in fact table
- Excluded 2 non-INR transactions due to lack of exchange rate reference.
- Reviewed negative and zero sales values for anomalies
- Standardized currency values
- Ensured relational integrity across dimension tables
- Verified no orphan foreign key records
- Confirmed one-to-many relationships integrity

4. Dataset Scope & Core Metrics

1. Time Period: 16/05/2018 – 28/05/2020
2. Total Orders: 1,47,061
3. Unique Customers: 77,584
4. Total Revenue: ₹ 96,42,58,527
5. Average Order Value (Mean): ₹6,557
6. Median Order Value: ₹519
7. Orders per Customer: 1.92

Total Orders	Avg AOV	Total Revenue
147,061	₹6,557	₹964,258,527

<u>Interpretation:</u>

- Mean > Median - Strong right skew
- Revenue influenced by premium transactions
- Majority of customers transact only once or twice

5. Exploratory Data Analysis (Excel)

5.1 Revenue Distribution & Outlier Analysis (IQR Method)

1. Q1 = ₹176
2. Q3 = ₹3,065
3. IQR = ₹2,889
4. Upper Threshold (1.5×IQR) = ₹7,398.50
5. Upper Outliers = 22,272 transactions

<u>Findings:</u>

- No lower-bound outliers
- Significant upper-tail concentration
- 80.75% of transactions fall under ₹5,000
- High-value transactions represent small volume but strong revenue contribution

<u>Interpretation:</u>

Revenue concentration is demand-driven, not supply-driven.

5.2 Customer Behaviour Analysis

<u>Orders per Customer Distribution:</u>

- 43.39% placed 1 order
- 32.63% placed 2 orders
- 76% placed only 1–2 orders
- High-frequency customers (>5 orders) <2%

<u>This indicates:</u>

- Low repeat depth
- Retention opportunity
- Engagement is wide but shallow

5.3 Restaurant-Level Revenue Concentration

- Top 10 restaurants contribute ₹1.36 Cr
- Less than 1% of total revenue
- Despite transaction-level concentration, restaurant-level revenue is dispersed.

<u>Conclusion:</u>

Revenue dominance is not caused by a few large restaurants.

6. Hypothesis Testing

H1: Income–Affordability Mismatch Impacts AOV Across Cities

Hypothesis:

Cities where restaurant cost exceeds income show lower AOV.

<u>Method:</u>

Pivot analysis by City × Income Band using AOV.

Average Order Value (AOV)	Income Grou					
City	10001 to 25000	25001 to 50000	Below Rs.10000	More than 50000	No Income	City Average AOV
Tirupati	166927	195625	235819	196005	155492	175634
Sirsi		306	110819	262278	132270	153836
Ujjain	18256	5812	11437	17381	7955	10067
Guna	4388	10505	6558	17774	10005	10000
Kammanahalli/Kalyan Nagar,Bangalore	14852	9659	7481	8570	9758	9990
Aligarh	5047	20107	1760	12871	7795	9933
Kumta					80	80
Naharlagun					79	79

Findings:

- In mid-range cities, higher income groups show higher AOV.
- In some high-AOV cities, lower income groups show equal or higher AOV.
- Relationship inconsistent across cities.

Conclusion:

Not fully supported.

Business Interpretation:

Income influences spending in some cases but does not structurally explain AOV variation.

Other factors likely dominate:

- Restaurant positioning
- Local demand behaviour
- Occasion-based ordering

H2: Order Volume Concentration Does Not Indicate Weak Revenue Quality

Method:

Pivot by Income Group with:

- Total Orders
- AOV
- Total Users

Income Group	Total Orders	Average Order Value (AOV)	Total Users
10001 to 25000	16910	6241.45	16910
25001 to 50000	26140	6529.65	26140
Below Rs.10000	9491	6995.42	9491
More than 50000	23687	6679.50	23687
No Income	70833	6542.43	70833

Findings:

- Mid-income (₹25k–50k) & High-income (>₹50k) drive highest volume
- AOV relatively stable across income groups
- No sharp AOV decline in lower-income segments

Conclusion:

Supported.

Implication:

- High volume in certain segments does not dilute revenue quality.
- Revenue monetization is balanced across income groups.

H4: Student Segment Dominates Order Volume

Orders by Occupation:

- Student: 78,301
- Employee: 44,830
- Self-employed: 20,541
- Housewife: 3,389

Occupatio	Total Orders
Employee	44830
House wife	3389
Self Employeed	20541
Student	78301

Conclusion:

Strongly supported.

Business Risk:

Platform heavily dependent on student segment.

Strategic implication:

Growth diversification required to reduce cohort concentration risk.

7. Tableau Visualization Insights

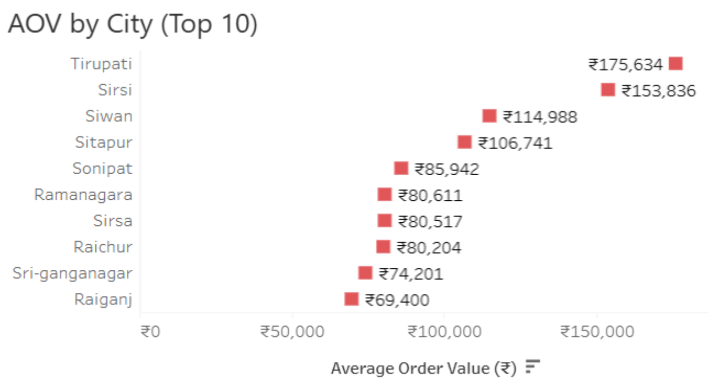
7.1 AOV by City (Top 10)

Significant variation:

₹69,400 – ₹175,634

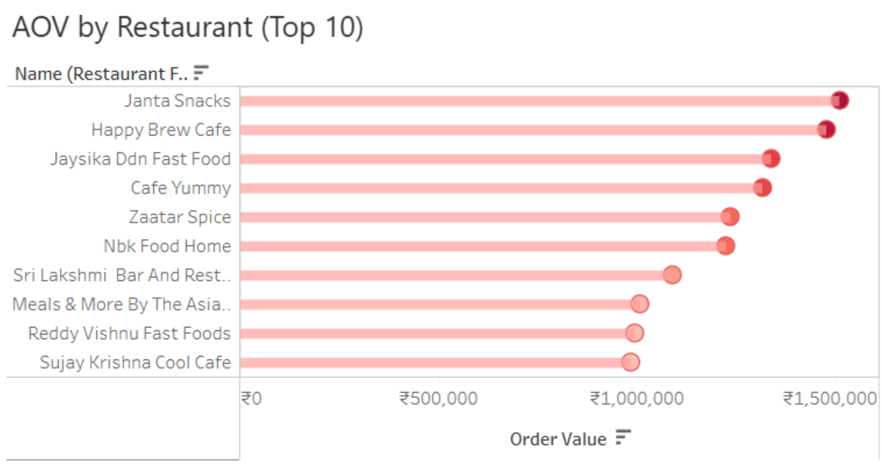
Indicates:

- Geographic monetization imbalance.



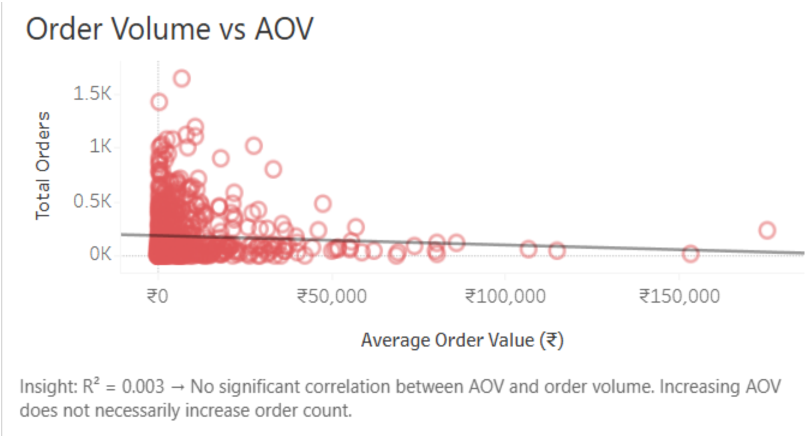
7.2 AOV by Restaurant (Top 10)

- High dispersion in restaurant-level order value.
- Not supply constrained.



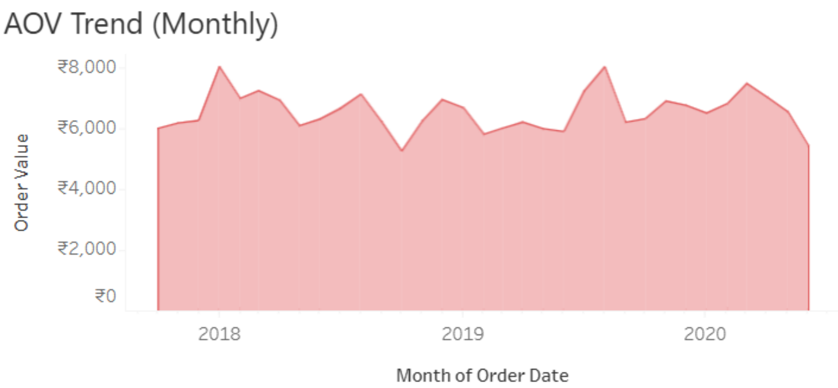
7.3 Order Volume Vs AOV

- $R^2 = 0.003$
- No meaningful correlation.
- Increasing AOV does not automatically increase order count.
- Revenue quality and order frequency behave independently.



7.4 Monthly AOV Trend

- Stable band between ₹6,000–₹8,000.
- No structural collapse.
- 2020 dip influenced by incomplete year data.



8. Underperforming Segments Identified

1. Low-frequency customers (1–2 orders)
2. Cities with low AOV and weak income linkage
3. Non-student occupation segments
4. Heavy reliance on upper-tail high-value transactions

9. Data-Backed Recommendations

9.1 Improve Repeat Behaviour

Problem:

76% customers transact only once or twice.

Action:

- Loyalty tiers for 3rd+ order
- Time-bound reorder incentives
- Subscription bundles

Data Justification:

Orders per customer = 1.90

9.2 Reduce Revenue Concentration Risk

Problem:

Revenue skewed toward upper-tail transactions.

- Action:
- Bundle pricing to increase mid-tier AOV
- Encourage add-ons
- Introduce “Smart Upsell” recommendations

Justification:

80% transactions under ₹5,000 heavy right skew.

9.3 Diversify Beyond Student Segment

Problem:

Students drive majority of volume.

Action:

- Corporate lunch partnerships
- Family-focused weekend promotions
- Targeted marketing for employees

Justification:

79k student orders vs 45k employees.

9.4 City-Level Pricing Optimization

Problem:

Since income does not consistently explain AOV

Action:

- Evaluate restaurant mix per city
- Improve assortment in low-AOV cities
- Local demand analysis

10. Final Conclusion

Zomato's platform demonstrates strong transaction activity and revenue generation. However:

- Revenue is highly concentrated in high-value transactions.
- Repeat customer depth is weak.
- Student segment drives disproportionate volume.
- AOV does not scale with order frequency.
- Income is not a consistent structural driver of AOV differences.

The business opportunity lies not in increasing volume alone, but in:

- Improving retention depth
- Enhancing mid-tier monetization
- Diversifying customer dependence
- Optimizing city-level revenue quality

This analysis combines Excel-based data validation and statistical exploration with Tableau-driven visualization insights to deliver a comprehensive evaluation of revenue efficiency and platform sustainability.